

Naming Hydrocarbons (using the IUPAC system)

IUPAC System: International Union of Pure and Applied Chemistry

Rules for naming Aliphatic Alkanes:

1. Find the **longest chain of carbon atoms** in the molecule ... STEM
2. Choose the **PREFIX** that corresponds to that length of chain (eg Meth- Eth- Prop- Pent- Hex- Hept-)
3. Look to see if there are **any substituent ALKYL chains** on the main chain:

Nature of Alkyl Group	Name of Alkyl group
-CH ₃	Methyl
-C ₂ H ₅ (or -CH ₂ CH ₃)	Ethyl
-C ₃ H ₈ (or -CH ₂ CH ₂ CH ₃)	Propyl

4. Identify the **position of the alkyl group** on the Carbon chain so that it gives the name with the **lowest number**
5. Use **di- ; tri- ; tetra- ;** etc to indicate **how many of each kind of substituent group** is present
6. **Where more than one substituent group are present, arrange them in alphabetical order**

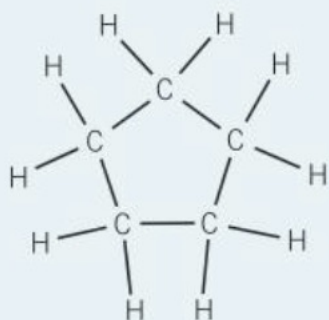
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Rules for naming Alicyclic Alkanes:

The same rules will apply as with Alkanes:

- The prefix “Cyclo-” is added to indicate that the carbon atoms in the alkane are arranged in a ring



▲ **Figure 4** *Cyclopentane*



Worked example: Naming alicyclic alkanes

How do you work out the IUPAC name of the compound in Figure 4?

Step 1: Identify the longest continuous chain of carbon atoms.

There are five carbon atoms in the ring structure; the stem is pent-.

Step 2: Add the prefix cyclo-.

The compound is called cyclopentane.

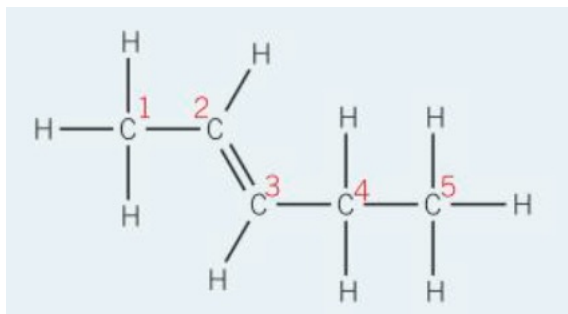
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Rules for naming Alkenes:

The same rules will apply as with Alkanes:

- The suffix used is “-ene”
- If the longest carbon chain has 4 or more carbon atoms, the position of the carbon-to-carbon double bond must be indicated



- SUFFIX is “-ene” as there is a carbon-to carbon double bond
- Longest chain – 5 so the STEM is “pent-”
- Carbon-to carbon double bond is between C2 and C3. We use the smallest of these numbers to indicate the position of the double bond, so -2.
- Combine the STEM, SUFFIX AND DOUBLE BOND POSITION TO NAME THE COMPOUND AS:

Pent-2-ene

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Rules for naming hydrocarbons containing Functional Groups:

The same rules will apply as with Alkanes:

- The STEM name comes from the corresponding Alkane (eg meth-, eth-)
- Identify any functional groups and any alkyl chains and the associated prefixes or suffixes that go with them
- Number the alkyl groups and functional groups to indicate their position on the longest unbranched chain



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